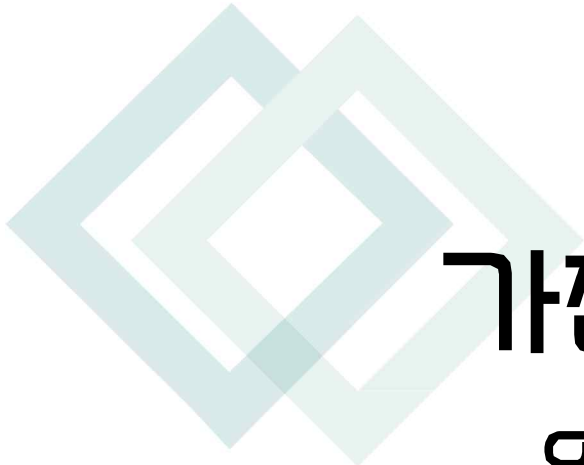




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연합 학술제

연구 파트 보고서

**Assessing Supply Chain Vulnerability and
Price Volatility in the Neodymium Market :
Evidence from Machine Learning Models**

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Abstract

Neodymium (Nd) price dynamics around COVID-19 reflect an interplay between *structural drivers* (production, demand, trade) and *event shocks* (policy/sanctions). Using ~ 10 years of monthly data, we estimate 84 tree-ensemble models across three cross-regime designs—Pre $\rightarrow$ COVID (R1), COVID $\rightarrow$ Recovery (R2), and Pre+COVID $\rightarrow$ Recovery (R3)—with baseline and leave-one-out sets. A news-based event score with month-end exponential decay (carry-over) is incorporated, and 12-month quantile scenario forecasts (Intensification / Maintenance / Relaxation; seed = 42) are produced. Results show structural-break generalization failure in R1 (best MAPE $\approx 51\%$); in R2, removing the event variable uniquely restores out-of-sample accuracy (LOO_Event MAPE $\approx 15\%$); in R3, retaining all covariates performs best (MAPE $\approx 9\%$), with the wind-energy index emerging as a key medium-term demand proxy, while the event score serves better as a scenario lever than as a core predictor. Scenario paths align for $\sim 3\text{--}6$ months before diverging, with average levels *Maintenance* $>$ *Relaxation* $>$ *Intensification*. Overall, structure dominates in normal times, event signals are best reserved for stress testing, and policy efficacy hinges on persistence (about 6–9 months) rather than one-off announcements.

1 Introduction

Neodymium, one of the rare earth elements (REEs), is an indispensable resource for advanced industries. It is widely used in electric vehicle motors, wind turbines, smartphones, and hard disk drives, making it a cornerstone of modern technological development (국제무역통상연구원, 2023). In particular, the importance of neodymium has been growing due to the rapid expansion of the electric vehicle market and the renewable energy industry. This is because neodymium is the primary raw material for neodymium-iron-boron (NdFeB) magnets—the strongest permanent magnets—used as critical components across a wide range of advanced technologies. Global consumption of NdFeB magnets increased by 13.3% from 2023 to 2024, and demand is projected to grow at an annual average rate of 8.7% between 2024 and 2040. This trend indicates a corresponding surge in global demand for neodymium (Intelligence, 2024).

On the basis of this industrial significance, neodymium has emerged as a key resource in economic competition among nations. However, the global supply chain of rare earths is structurally fragile and highly concentrated in China (고윤성, 2025). As of 2023–2025, China accounts for approximately 60% of global REE production, while 89% of refining and processing also takes place in China (U.S. Geological Survey, 2025). Moreover, China controls about 92% of global permanent magnet production (국제무역통상연구원, 2023). Such structural dependence on a single country not only raises concerns about resource availability but also poses significant risks in terms of national security and industrial strategy. Therefore, this study aims to analyze the neodymium market from a global perspective by constructing a composite dataset that integrates historical price and industrial market data, rare earth-related policies, tariff regulations, and event-based information to provide a more comprehensive understanding of market dynamics.

Existing domestic and international studies have often focused on limited variables such as national policies, reserves, or stock prices, or have relied solely on single time-series analyses (El Azhari et al., 2024; Henriques & Sadorsky, 2023; 오정미, 2024). Extending this line of research, Gupta proposed a Regularized Sparse Autoencoder (RSAE)-based prediction model that integrates multiple time-series inputs, including copper and crude oil futures prices, while incorporating macroeconomic indicators and supply-

demand factors (Gupta, 2025). This approach enhanced both predictive accuracy and interpretability.

While rare earths are increasingly recognized as strategic resources in advanced industries and energy transitions, quantitative research that systematically integrates external shocks and market data for specific rare earth elements remains limited. In particular, studies addressing metals such as neodymium—characterized by both high supply concentration and geopolitical risks—are rare. Research that combines policy variables, external events, and China’s dominant supply chain structure into an integrated quantitative framework is still lacking.

This study addresses this gap by analyzing the structural characteristics of the neodymium supply chain and its sensitivity to external shocks. In doing so, it aims to contribute to long-term resource security strategies and industrial policy planning. To this end, we construct an empirical dataset covering the period from 2015 to 2025, including variables such as prices, production, demand, and trade flows, along with major events such as the COVID-19 pandemic and geopolitical conflicts. Given China’s dominance in refining and metallization stages, quantitatively identifying the volatility and shock responses of the neodymium market provides essential insights for enhancing industrial competitiveness and formulating effective crisis management strategies.

Through this approach, the study offers a quantitative assessment of structural changes and shock responses in the neodymium market before and after COVID-19. It examines the interrelationships among production, demand, price indices, and external events in depth, and evaluates the impacts of major disruptions—such as pandemics, geopolitical conflicts, and policy changes—on market volatility. Furthermore, by comparing structural dependence during normal periods with adjustment capacity during crises across different scenarios, this study provides a multidimensional analysis of global supply chain risks. Ultimately, it seeks to derive policy implications that encompass both structural dependence management and shock-response strategies, thereby informing future resource security and trade policy.

To achieve this, the study addresses the following four key research questions:

- RQ1. *Structural evolution*: Characterize how structural dynamics and shock responses in the neodymium market changed from the pre-COVID period to the post-COVID recovery.
- RQ2. *Drivers of price variation*: Quantify the respective contributions of production, demand, and event-related variables to price fluctuations.
- RQ3. *Event impacts and resilience*: Estimate the volatility effects of major events (pandemic, geopolitical tensions, policy shifts) and infer market vulnerability and resilience.
- RQ4. *Regime contrasts and risk*: Compare structural dependence in normal times with adjustment capacity during crises across scenarios, and map the implications for global supply-chain risk.

This research design contributes to the literature by integrating policy scenarios, long-term time-series data, and external shock events, thereby enabling a comprehensive analysis of structural risks in the neodymium market. Nonetheless, certain limitations must be acknowledged.

First, the analysis is based on data from August 2015 to July 2025; therefore, abrupt policy changes, technological breakthroughs, or geopolitical risks beyond this period may not be fully captured. Second, the predictive model is path-dependent, based on current assumptions, and may not reflect complex and unforeseen developments beyond the scenarios examined in this study. Third, variables originating from countries other than China—such as policies, technological innovations, and investment trends—were not considered.

These limitations suggest several directions for future research. Subsequent studies could expand the scope by quantitatively assessing the effectiveness of rare earth self-sufficiency strategies pursued by countries outside China. Sensitivity analyses of policy changes, such as variations in export controls or the tightening of carbon regulations, could also be conducted. Finally, refining demand scenarios for key technologies—such as wind power generation and electronic devices—could further improve the precision of predictive models.

2 Methodology

2.1 Forecasting Directions

To account for the structural disruption caused by the COVID-19 pandemic, this study designs three distinct forecasting directions. First, the *Pre*→*COVID* setting employs data from the pre-pandemic period as the training set and the pandemic impact phase as the validation set. This design evaluates the generalization capability of models when exposed to unprecedented structural breaks, with the expectation that forecasting errors may be magnified. Second, the *COVID*→*Recovery* setting uses the pandemic impact phase for training and the recovery phase for validation. This approach highlights whether models can internalize pandemic dynamics and subsequently generate reliable forecasts during the recovery stage. Finally, the *Pre*+*COVID*→*Recovery* direction combines both pre-pandemic and pandemic impact data as the training set, while the recovery phase is used for validation. From a practical perspective, this scenario is expected to provide the most stable and accurate forecasts by exploiting information across multiple regimes.

2.2 Variable Sets

Seven sets of explanatory variables are constructed to examine the robustness of forecasting performance. The baseline includes a univariate model with only the price series (*CO*) and a comprehensive model incorporating all available covariates (*Call*), such as production volumes, demand indicators, policy events, substitute metals, and exchange rates. To evaluate the marginal contribution of each covariate group, a leave-one-out (LOO) strategy is applied to the *Call* set. Specifically, production variables (LOO_w), demand measures (LOO_m), policy events (LOO_p), substitutes (LOO_n), and exchange rates (LOO_e) are excluded in turn, thereby allowing the incremental role of each group to be assessed.

2.3 Algorithms

Each variable set is evaluated through four tree-based ensemble algorithms: Random Forest (RF), XGBoost (XGB), LightGBM (LGBM), and CatBoost (CAT). These algorithms are chosen for three primary reasons. First, they are effective in relatively short time series, such as monthly data of approximately 120 observations. Second, they are well suited to capturing nonlinearities and complex interactions among covariates. Third, they provide interpretability through feature importance measures and systematic exclusion (LOO), which is particularly advantageous for drawing policy implications.

2.4 Experimental Scope

The full experimental design consists of 3 forecasting directions, 7 variable sets, and 4 algorithms, resulting in a total of 84 models. This structure allows systematic comparison across three dimensions: (i) the generalization ability of models under structural disruptions, (ii) the relative contribution of different covariates, and (iii) forecasting accuracy from a practical standpoint.

2.5 Evaluation Metrics

Forecast performance is evaluated using four complementary error measures. Mean Absolute Error (MAE) captures the average magnitude of errors in an intuitive manner, while Root Mean Squared Error (RMSE) penalizes large deviations more heavily, thereby emphasizing stability. Mean Absolute Percentage Error (MAPE) enables comparison in relative terms by scaling errors to predicted values. Finally, Mean Absolute Scaled Error (MASE) provides a normalized metric relative to a naive benchmark, facilitating comparative assessment across models. Together, these metrics offer a balanced evaluation framework that accounts for absolute error size, relative interpretability, and robustness.

3 Data and Descriptive Statistics

3.1 Data

In this study, we analyze the volatility of the neodymium market before and after COVID-19, using four main datasets: price data, global production data, supply shock and event data, and demand data for the neodymium market. The dataset covers the period from August 2015 to July 2025, and all data were converted into monthly frequency for analysis.

Production data were based on official Chinese customs statistics for neodymium metal exports, and global export volumes were estimated using China's market share as a benchmark.

Table 1:
Estimated market shares of Magnet REEs and refined Nd with sources

Year	Magnet REEs Refined Share	Conversion Ratio	Nd Refined Share	Source
2015	88.5% (est.)	$\times 0.935$	82.8%	Interpolated estimate
2016	88.5% (est.)	$\times 0.935$	82.8%	Interpolated estimate
2017	89.0% (est.)	$\times 0.935$	83.3%	Interpolated estimate
2018	89.5% (est.)	$\times 0.935$	83.7%	Interpolated estimate
2019	89.7% (est.)	$\times 0.935$	83.8%	Interpolated estimate
2020	90.0% (confirmed)	$\times 0.935$	84.2%	Converted
2021	89.8% (confirmed)	$\times 0.935$	84.0%	Actual figure confirmed
2022	91.0% (est.)	$\times 0.935$	85.1%	Interpolated estimate
2023	92.1% (confirmed)	$\times 0.935$	86.1%	Converted
2024	92.3% (est.)	$\times 0.935$	86.3%	Interpolated estimate
2025	92.5% (est.)	$\times 0.935$	86.5%	Interpolated estimate

However, this study assumes refined metal quantities as production volumes and uses national export data as a proxy indicator. As a result, domestic sales in each country are not fully reflected, which constitutes a limitation of the dataset.

Table 2:
Example of five news articles

Title	Direction	Magnitude	Channel	Continuity	confidence
중국의 보복...	1	3	direct	60	0.9
폐기된 영구자석에서...	0	1	indirect	7	0.3
무역전쟁의 ‘카드’ ...	1	3	direct	30	0.9
중국 최대 광산 지역...	0	1	indirect	7	0.3
중국의 북방회토...	1	3	direct	60	0.9

News events were classified according to their content. Articles mentioning specific items or policy signals such as neodymium (Nd), praseodymium (Pr), NdFeB, permanent magnets, export controls, quotas, or production cuts were categorized as *direct* channels (=1). Articles addressing indirect factors such as electric vehicle and wind power demand, substitution technologies, or macroeconomic trends were categorized as *indirect* channels (=0). The price impact direction was classified as upward (+1), downward (−1), or neutral (0). The magnitude of impact was categorized into three levels: 1 (weak), 2 (moderate), and 3 (strong). The duration of impact was defined by half-life periods (7, 14, 30, 60 days) as *continuity*. Finally, a *confidence* score between 0 and 1 was assigned based on a composite evaluation of the news agency’s reliability, certainty of the wording, and specificity of the item mentioned.

3.2 News-based Event Score with Monthly Carry-over

We construct a news-derived event measure that aggregates signed shocks at a monthly frequency while allowing effects to decay exponentially across month-ends and to expire once negligible. For an article i , the raw signal combines direction and a smoothed intensity:

$$S_i = \text{direction}_i \times f(\text{magnitude}_i), \quad f(m) = \log(1 + m). \quad (1)$$

Channel and credibility adjust the signal through a multiplicative weight:

$$W_i = g(\text{channel}_i) \times \text{confidence}_i, \quad g(c) = 1 + \alpha c, \quad c \in \{0, 1\}, \alpha \in [0, 1]. \quad (2)$$

The initial contribution is given by

$$C_i = S_i \times W_i. \quad (3)$$

Assuming exponential decay, article i with half-life τ_i has decay rate $\lambda_i = \ln(2)/\tau_i$. Let t_m denote the last day of month m and d_i the article date. The residual contribution evaluated at month-end t_m is

$$D_{i,m} = C_i \exp(-\lambda_i \Delta t_{i,m}), \quad \Delta t_{i,m} = \max\{0, (t_m - d_i)\} \text{ days}. \quad (4)$$

Contributions smaller than one percent of the initial value are discarded thereafter. Equivalently,

$$D_{i,m} = \begin{cases} C_i \exp(-\lambda_i \Delta t_{i,m}), & \exp(-\lambda_i \Delta t_{i,m}) \geq \varepsilon, \\ 0, & \text{otherwise,} \end{cases} \quad \varepsilon = 0.01, \quad (5)$$

so carry-over persists only while $\Delta t_{i,m} \leq \lambda_i^{-1} \ln(1/\varepsilon)$.

The monthly news shock index is the month-end sum of all valid residuals:

$$M_m = \sum_i D_{i,m}. \quad (6)$$

To render the series comparable across the sample, we standardize by the full-sample mean μ and standard deviation σ :

$$Z_m = \frac{M_m - \mu}{\sigma}. \quad (7)$$

With $\tau_i = 30$ days, $\lambda_i = \ln(2)/30 \approx 0.0231$. If an event on 10 Aug 2025 is evaluated at 31 Aug 2025, then $\Delta t_{i,m} = 21$ and

$$\exp(-\lambda_i \Delta t_{i,m}) \approx \exp(-0.0231 \times 21) \approx 0.615,$$

so an initial $C_i = 10$ implies $D_{i,m} \approx 6.15$. The one-percent cutoff implies expiry after roughly

$$\Delta t > \frac{\ln(100)}{\lambda_i} \approx \frac{4.605}{0.0231} \approx 199 \text{ days.}$$

Neodymium is not directly consumed as a final product but is primarily used as a key raw material in downstream industries, including rare earth permanent magnets (NdFeB), wind turbines, and electric vehicle drive motors. Accordingly, this study indirectly estimates neodymium demand by employing indicators from three representative industries, focusing on permanent magnet and wind power indices, while using new energy vehicle (NEV) data as a supplementary indicator.

3.3 Descriptive Statistics

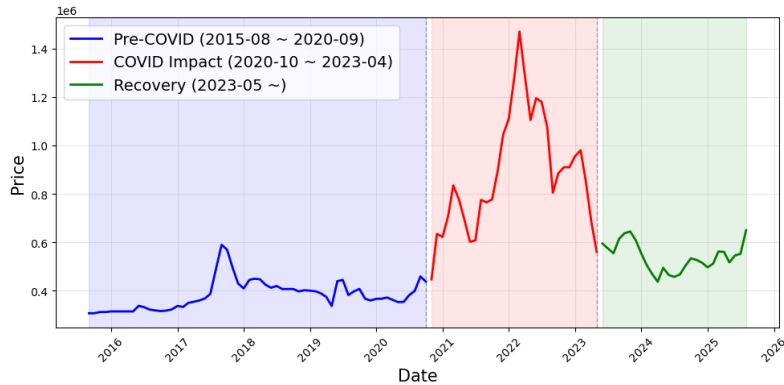


Figure 1: Neodymium Price Graph by Period

To establish a robust foundation for the subsequent analysis, we first present the descriptive statistics of the key variables across the pre-COVID, COVID-shock, and recovery periods. These statistics provide an

overview of the central tendency, dispersion, and distributional features of the data, thereby allowing us to identify structural differences in the neodymium market over time. By reporting measures such as mean, variance, skewness, and kurtosis, we capture both the average market conditions and the extent of volatility and asymmetry that characterize the distribution of prices, production, demand, and event-driven indices. This descriptive step is essential, as it enables us to trace how the pandemic, geopolitical tensions, and policy shifts altered the statistical properties of the market. In doing so, it lays the groundwork for evaluating the market's vulnerability to shocks and its subsequent resilience, which directly informs our broader goal of assessing structural dependence and adjustment capacity within the global supply chain.

Table 3:
Descriptive Statistics Before COVID-19.

	Price	Production	NEV sales	Wind-energy index	Magnet exports	Event score
count	62	62	62	62	62	62
mean	382960	51570	123788	12.89	3.78e+08	-0.307
std	60460.48	28818.40	63444.83	1.34	6.42e+07	0.565
min	307500	238	26292	10.47	1.77e+08	-0.779
25%	334500	29749	75572	12.12	3.57e+08	-0.725
50%	373750	47548	119095	12.94	3.96e+08	-0.587
75%	409375	77397	156253	13.29	4.21e+08	0.009
max	590000	103823	316111	17.38	4.70e+08	1.468
Skewness	1.136	0.131	0.639	1.087	-1.404	1.486
Kurtosis	1.756	-1.061	0.057	2.509	1.695	1.504

Table 4:
Descriptive Statistics During COVID-19

	price	production	nev-sales	wind-energy-index	magnet-exports	event-score
count	31	31	31	31	31	31
mean	885000	46481	713528	19.60	6.09e+08	0.385
std	242825	31865	278485	2.23	1.06e+08	0.917
min	447000	594	271044	15.33	4.13e+08	-0.841
25%	702500	23901	510947	17.82	5.50e+08	-0.394
50%	855000	38190	668050	18.93	6.24e+08	0.291
75%	1060000	75060	898209	21.37	6.82e+08	0.806
max	1470000	108696	1340444	24.76	7.99e+08	2.992
Skewness	0.457	0.378	0.379	0.331	-0.200	0.747
Kurtosis	-0.382	-1.092	-0.716	-0.610	-0.659	0.373

Table 4 shows that during the COVID-19 period, market variables displayed high volatility, with price and production exhibiting large dispersions. Table 5 highlights the recovery stage, where both price and production stabilized at lower average levels, while sales began to regain momentum.

Table 5:
Descriptive Statistics Recovery Period

	price	production	nev-sales	wind-energy-index	magnet-exports	event-score
count	27	27	27	27	27	27
mean	539444	25949	1496599	16.18	4.24e+08	0.478
std	58993.05	30310.41	301460	1.25	1.79e+08	1.665
min	437500	35	854187	13.74	8.12e+07	-1.190
25%	500500	2771	1300110	15.25	3.09e+08	-0.652
50%	534000	15100	1391697	15.92	5.07e+08	-0.588
75%	568500	38898	1774443	17.06	5.33e+08	1.488
max	650000	131678	2083861	19.13	6.23e+08	4.788
Skewness	0.282	1.770	0.110	0.327	-0.976	1.288
Kurtosis	-0.757	3.633	-0.690	-0.441	-0.740	0.581

Table 6:
Descriptive Statistics (Total Data)

	price	production	nev-sales	wind-energy-index	magnet-exports	event-score
count	120	120	120	120	120	120
mean	547863	44491	585020	15.36	4.48e+08	0.048
std	247573	31425.73	587495	3.25	1.46e+08	1.058
min	307500	35	26292	10.47	8.12e+07	-1.190
25%	371500	21904	116318	12.92	3.66e+08	-0.693
50%	458500	38020	249352	14.81	4.29e+08	-0.359
75%	616875	72903	1054836	17.51	5.39e+08	0.421
max	1470000	131678	2083861	24.76	7.99e+08	4.788
Skewness	1.555	0.432	0.906	0.716	-0.044	2.020
Kurtosis	1.981	-0.732	-0.507	-0.258	0.178	4.861

Table 6 summarizes the full-sample statistics. Compared with the period-specific results, the total data reveal persistent volatility in prices and exports, confirming that neodymium market dynamics remained sensitive to shocks across the entire observation window.

As shown in Fig. 2, the correlation matrices for neodymium-related variables are reported for four periods: Pre-COVID, COVID Impact, Recovery, and Total. In the Pre-COVID period, price exhibits moderate positive correlations with magnet exports ($r \approx 0.53$) and the wind-energy index ($r \approx 0.44$), suggesting a demand-driven structure. During COVID, price correlations shift toward demand-side indicators such as NEV sales ($r \approx 0.29$) and the wind-energy index ($r \approx -0.35$), while event shocks are negatively associated ($r \approx -0.40$). In the Recovery phase, price shows a stronger association with production ($r \approx 0.52$), indicating the re-emergence of supply-side effects. For the total sample, price is most strongly correlated with the wind-energy index ($r \approx 0.69$) and magnet exports ($r \approx 0.68$), highlighting the structural importance of renewable-energy demand and international trade in long-run price dynamics.

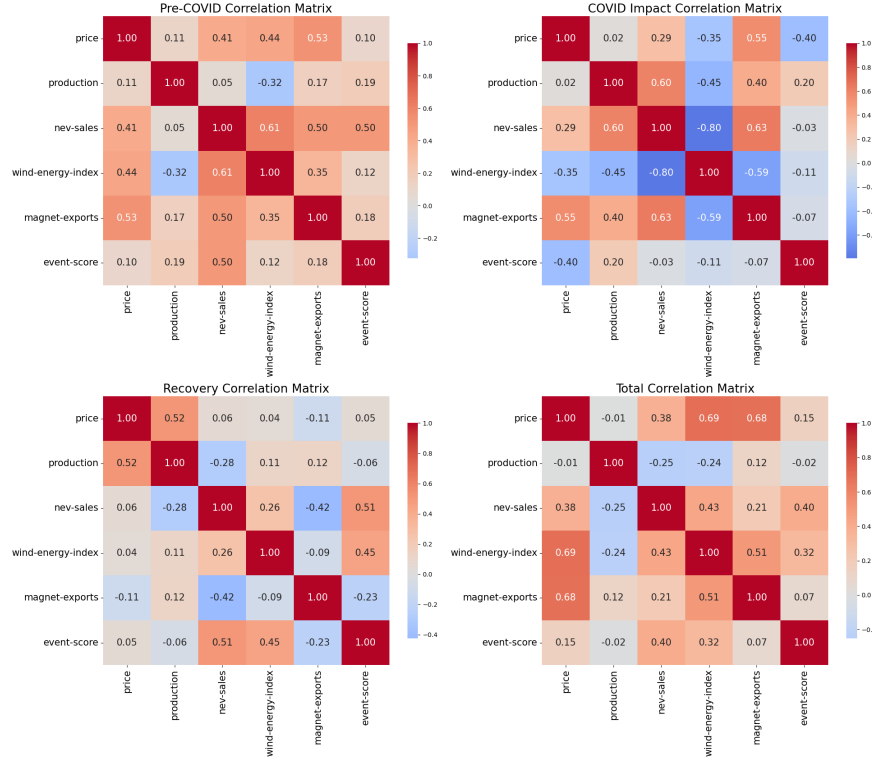


Figure 2: Correlation Matrix by Period

4 Results

4.1 Scenario-Level Summary

We evaluate 84 model configurations spanning three forecasting directions (R1–R3), seven variable sets (univariate price, all covariates, and five leave-one-out variants), and four ensemble learners (RF, XGB, LGBM, CAT). For concision, we report the scenario-defining outcomes that governed qualitative conclusions in the main text and relegate the full grid of results to the appendix. Throughout, we emphasize MAPE for comparability across regimes; companion MAE, RMSE, and MASE tables are provided in the supplementary material.

Table 7:
Scenario-wise minimal MAPE configurations.

Scenario (train → test)	Algorithm	Configuration	MAPE (%)
R1: Pre→COVID	XGBoost	LOO_NEV	50.77
	XGBoost	All covariates †	52.63
	XGBoost	LOO_Event	53.90
R2: COVID→Recovery	XGBoost	LOO_Event †	15.14
	XGBoost	All covariates	64.75
R3: Pre+COVID→Recovery	XGBoost	All covariates	9.24
	XGBoost	LOO_Production †	9.39

Notes. Bold numerals indicate scenario-wise minima. The † symbol (†) marks the author-selected primary specification based on trajectory fidelity and robustness, which may differ from the minima. Boldface in configuration names is a visual cue only and has no inferential meaning.

As summarized in Table 7, the scenario-wise selections reflect both numerical fit and trajectory fidelity. In R1 (Pre→COVID), *LOO_NEV* attains the lowest MAPE (50.77%), whereas the *All covariates*[†] specification is retained as a qualitative benchmark because it more faithfully tracks the sharp run-up of 2021–2022 and subsequent inflections; *LOO_Event* is competitive and exhibits close path adherence. In R2 (COVID→Recovery), excluding the event score is decisive: most alternatives exceed 60% MAPE, while *LOO_Event*[†] generalizes best at 15.14%, indicating that rules learned on shock-period headlines do not transport well into the normalization regime. In R3 (Pre+COVID→Recovery), *All covariates* delivers the lowest error (9.24%), whereas *LOO_Production*[†]—despite a slightly higher MAPE (9.39%)—most closely reproduces the recovery-phase trajectory, capturing level shifts and turning points. Taken together, these patterns are consistent with the broader evidence that regime-diverse training anchored in structural fundamentals yields the most reliable forecasts, while episodic announcements are better suited to scenario design than to routine point prediction.

4.2 R1: Training on Pre-Pandemic Data and Validating in the Shock Phase

Across specifications trained exclusively on the pre-pandemic regime, forecast accuracy deteriorates markedly when confronted with the pandemic impact period. The pattern reflects a structural-break penalty: abrupt level shifts and volatility bursts are not inferable from stable-history splits, and the addition of standard covariates does not bridge the regime gap. The slight gain from removing NEV proxies suggests that strong trend and level effects embedded in NEV adoption can steer tree partitions toward regime-specific splits that fail in collapse segments; eliminating the offending proxy attenuates this fragility. That feature importance occasionally ranks production or NEV lags highly in R1 does not contradict this interpretation, as impurity-based importance is conditional on the realized partition, whereas the leave-one-out exercise measures marginal contribution under re-optimization and can reveal harmful variables under collinearity.

4.3 R2: Training in the Shock Phase and Validating in the Recovery

When models are trained on the pandemic impact period, event-score lags dominate split selection and deliver apparently strong in-sample fit; however, once the system normalizes, these rules fail to reproduce, and generalization improves precisely when the event-score is omitted. Removing events forces the learner to refocus on production, trade, and demand indicators that carry through to the recovery regime. The divergence between high importance of specific event lags and their negative marginal contribution is a textbook instance of conditional-versus-marginal contrast in tree ensembles under regime change.

4.4 R3: Joint Training on Pre-Pandemic and Shock Phases and Validating in the Recovery

Combining pre-pandemic and pandemic data in training yields the most balanced out-of-sample performance in the recovery, consistent with the value of heterogeneous-regime exposure for learning level transitions and volatility states. The wind-energy index is pivotal: removing it collapses performance, indicating that medium-horizon demand cycles tied to renewable deployment anchor the level and slope of prices. Event information remains useful in a supportive role—its removal modestly worsens accuracy—but the dominant drivers are structural fundamentals rather than episodic announcements.

4.5 Selected Quantitative Results (XGB)

In R1, the univariate price model attains 52.37% MAPE, the all-covariate model 52.63%, and the best-performing *LOO_NEV* 50.77%. In R2, the all-covariate configuration inflates error to 64.75%, while *LOO_Event* reduces it to 15.14%; some exclusions (e.g., magnet proxies) perform poorly, exceeding 100% MAPE. In R3, the all-covariate model achieves 9.24% MAPE; excluding wind raises error to 53.45%, whereas removing events, NEV, or magnet proxies yields intermediate degradations in the 11–14% range. Scenario-specific feature-importance visualizations are provided in the [Appendix B](#).

4.6 Twelve-Month Trade-Sanctions Scenario (*random_state* = 42; 80% Intervals)

To appraise policy-sensitive trajectories, we retain the effective R3 structure and fit an XGB specification that excludes production-side covariates in training to preserve parsimony and avoid leakage, while injecting a twelve-month event-score sequence at forecast time (stronger shocks in the first half, weaker thereafter). Covariates not directly shocked are extrapolated with univariate components consistent with in-sample dynamics. The forecast origin is July 2025 with a spot of 650,000. Table 8 summarizes median paths and average 80% interval widths.

Table 8:
Twelve-month sanctions scenarios (median).

Scenario	12M mean	Change	Final month	Avg. 80% CI width
Intensification	590,316	−9.2%	669,693	220,684
Maintenance	612,839	−5.7%	720,397	239,344
Relaxation	602,732	−7.3%	705,737	228,796

All scenarios co-move over the initial quarter as structural autocorrelation and fundamentals dominate, with troughs near 545–572k. Paths diverge from months six to nine as the interaction of event-score carry-over and covariate extrapolation widens. The maintenance case climbs to the highest terminal level, relaxation underperforms on average because attenuation of the shock lowers cumulative carry-over, and intensification remains persistently below both. Uncertainty bands average 220–240k; intervals overlap in the first half and separate thereafter, improving scenario discriminability in months seven to twelve.

4.7 Synthesis

Three messages emerge. First, structural breaks undermine the transportability of rules learned in tranquil regimes, and adding more covariates cannot fully compensate in shock periods. Second, episodic event information is powerful for fitting shocks but can degrade recovery-phase generalization unless tempered; excluding events in R2 restored focus on stable fundamentals. Third, regime-diverse training aligned with structural indicators—notably wind-related activity—yields realistic recovery forecasts with materially lower error and more credible policy simulations. Full grids of scores and importance plots are reported in the [Appendix B](#) for auditability.

5 Policy Implications

Evidence in [Section 4](#) supports a dual-track stance: reinforce structural fundamentals while compartmentalizing episodic shocks. The recommendations below retain the original substance in a more concise form.

5.1 Reinforcing Structural Fundamentals

Recovery-phase accuracy is maximized when models train across heterogeneous regimes and retain all covariates, with wind-sector activity a pivotal demand proxy. Policy should prioritize upstream processing/refining, recycling of end-of-life NdFeB, and allied capacity expansion; where rapid domestic ramp-up is infeasible, diversify via structured offtake and jurisdictional spread.

5.2 Treat Events as Scenario Levers, Not Core Predictors

Shock-phase fitting makes event indicators dominate splits but yields weak transportability into recovery. Operationally, isolate event information for stress tests and counterfactuals rather than day-to-day forecasting, preserving the stability granted by fundamentals while still exploring tail paths under sanctions, quotas, or licensing shocks.

5.3 Procurement and Hedging Across Horizons

Twelve-month scenarios show co-movement for ~3–6 months (structural autocorrelation) and divergence thereafter (event carry-over). In practice: hold 3–4 months of safety stock, calibrate partial hedges to forecast-band widths, and from month six condition contracting/hedging on shock persistence (e.g., widen forward coverage when intensification falls 5–7% below maintenance while wind/NEV momentum rises). Beyond one year, hedge structurally via supply-node diversification and recycling/joint investments.

5.4 Early Warning and Decision Routines

Adopt a lightweight dashboard with explicit triggers: wind/NEV for demand pressure, production/trade for supply tension, and a standardized event score with exponential carry-over for discrete risks. Review procurement calendars and hedge ratios when the 3-month event z -score breaches thresholds or when predictive bands widen materially; backtest quarterly against withheld periods and synthetic shocks.

5.5 Predictable Communication and International Coordination

Rules-based, pre-announced communication around controls, licensing, and permitting compresses uncertainty premia. Shared transparency standards (production/export statistics, outage reporting), interoperable provenance/recycled-content rules, and reciprocal access to strategic stocks temper cross-border amplification during stress episodes.

5.6 Implementation for Research and Practice

Deploy the regime-diverse training design that dominated in recovery, keep structural covariates as the forecasting backbone, and reserve the event index for scenarios/stress tests. Maintain documented data pipelines (wind, NEV, production, trade), refresh models on a rolling schedule that explicitly re-checks structural-break and shock-to-recovery splits, and publish the full grid of accuracy tables and feature diagnostics for replicability.

5.7 Synthesis

Short-horizon volatility remains announcement-sensitive, whereas medium-run levels are governed primarily by supply–demand fundamentals tied to the green-industry cycle. Policies that fortify fundamentals, manage information frictions through predictable communication, and institutionalize regime-aware forecasting with explicit stress routines are most likely to stabilize both price levels and variance while preserving flexibility under uncertainty.

6 Conclusion

This study quantified how neodymium prices reflect both structural dependence—rooted in production, demand, and the green-capex cycle—and sensitivity to episodic shocks arising from policy and geopolitical events. We organized the sample around three forecasting directions (R1: Pre→COVID; R2: COVID→Recovery; R3: Pre+COVID→Recovery) and evaluated seven variable sets across four tree-ensemble learners, yielding eighty-four models in total. To measure discrete shocks consistently, we developed a news-derived event index that combines logarithmic intensity, channel weighting, exponential month-end carry-over with expiry at a one-percent threshold, and full-sample standardization.

6.1 Main findings

Across regimes, four patterns emerge. First, in R1, models trained exclusively on the pre-pandemic period incur uniformly high errors when transported into the pandemic phase, indicating a structural-break penalty that cannot be neutralized by simply adding conventional covariates. Second, in R2, out-of-sample accuracy improves when the event indicator is excluded from the core predictor set, consistent with headline-driven partitions that fit the shock phase but fail to generalize once conditions normalize. Third, in R3, exposing learners to both tranquil and stressed distributions yields the most reliable recovery forecasts; retaining the full covariate set compresses errors materially, while the wind-sector proxy is pivotal for medium-horizon demand and level anchoring—its removal causes pronounced degradation. Fourth, twelve-month sanctions scenarios reveal a common short-run co-movement followed by divergence from months six to nine as event carry-over interacts with extrapolated fundamentals; the median level ordering favors persistent maintenance signals over transient relaxations, with intensification remaining lowest (see [Table 8](#)).

6.2 Contributions

Methodologically, the paper advances a regime-aware evaluation design that separates structural and episodic channels and compares them under explicit transport tests. The event index operationalizes shocks for scenario analysis by linking signed news to a transparent decay-and-expiry mechanism. The joint reading of impurity-based feature importance and leave-one-out re-optimization clarifies when variables that appear salient in-sample become harmful under distributional change. Practically, the results translate into an implementable rule set: retain structural covariates as the forecasting backbone, reserve events for stress testing and counterfactual paths, and couple forecasts to procurement and hedging routines that escalate with the persistence of shocks rather than their headlines.

6.3 Limitations and future work

The monthly sample is modest (on the order of one hundred twenty observations), several quantities rely on defensible proxies where direct measurements are unavailable, and the event index depends on a specific corpus and coding pipeline. Extrapolations for exogenous covariates are univariate, so joint uncertainty propagation is conservative. Future research can enrich the framework with regime-switching or state-space specifications and cointegration structures; broaden the shock pipeline using multilingual news and policy calendars to better assess persistence; and adopt time-series cross-validation with systematic hyperparameter search and permutation-/SHAP-based diagnostics to strengthen interpretability and robustness.

6.4 Implications

While short-horizon volatility remains sensitive to discrete announcements, medium-run trajectories are governed primarily by supply–demand fundamentals tied to green-industry demand. A dual-track strategy follows: manage structural dependence through diversified upstream capacity, recycling, and transparent stock policies, and control shock risk with scenario-based playbooks that emphasize persistence over one-off signals. The regime-aware forecasting and event-scoring architecture developed here provides a practical template for designing, auditing, and executing such a strategy in rare-earth markets.

A Data

The price data were obtained from Trading Economics, which provides weekly quotations for neodymium metal (China FOB, 99%) (Trading Economics, n.d.-a). In this study, metal prices were adopted instead of oxide prices because metal prices are more directly linked to final demand and therefore considered more appropriate for evaluating supply chain and industrial impacts. The reliability of the price data was cross-validated using the Shanghai Metals Market and the Argus Rare Earths Monthly Outlook (Argus Media, 2025; Metal.com, 2025).

The export volumes of countries other than China were interpolated and estimated on a yearly basis using data from the IEA Global Critical Minerals Outlook (2024), Rare Earth Permanent Magnets (2022), and Challenges and Opportunities in Global Supply Chains (2023). These were then combined with China's export volumes to calculate global exports. (Challenges and Opportunities in Global Supply Chains: The Role of Critical Minerals, 2023; Global Critical Minerals Outlook 2024, 2024; Rare Earth Permanent Magnets, 2022).

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This study constructed supply and demand shock indices to explain short-term fluctuations in neodymium prices, based on international trade-related news events. The event data were collected from the BIGKINDS news database using the keywords "rare earths" and "neodymium," and the Gemini-2.5-pro model was employed in the index construction process (Korea Press Foundation, n.d.). In cases where multiple media outlets reported on the same event, only one representative article was selected to remove duplication.

Demand for permanent magnets was derived from UN Comtrade, by aggregating the global monthly export volumes under HS850511 and HS850519 codes to build a time series from August 2015 to June 2025. Based on previous studies and official classifications, NdFeB magnets account for approximately 80–90% of the rare earth permanent magnet market, and they are classified under the above HS codes. Therefore, this was used as a proxy variable for neodymium demand (International Energy Agency, 2021; U.S. Customs and Border Protection, n.d. U.S. Department of Energy, 2023; 2024).

The wind energy index is an indicator that comprehensively reflects the performance of energy production and publicly traded wind-related companies, including firms that supply products and services to the wind industry and generate most of their revenues from it. In this study, the index provided by Trading Economics was employed for the analysis (Trading Economics, n.d.-b).

Since neodymium permanent magnets are essential components of NEV drive motors, the sales volumes of such vehicles can indirectly capture neodymium demand. Therefore, global monthly sales of EV, PHV, and FCV vehicles provided by MarkLines were utilized (MARKLINES, n.d.). In this study, NEV data were used as a supplementary indicator, while the main analysis focused on permanent magnets and the wind energy index.

B Extended Results: Complete Error Metrics by Scenario

Table 9:

Full results for R1 (Pre→COVID | train=*pre_covid*, test=*covid_impact*).

Algorithm	Spec	MAE	RMSE	MAPE	MASE
RF	C0	471831.45	530271.92	49.58 %	7.53
	Call	485866.45	543389.88	51.27 %	7.75
	LOO_Wind	479034.19	537264.46	50.46 %	7.64
	LOO_Magnet	485843.87	542656.94	51.32 %	7.75
	LOO_Production	494315.00	552388.18	52.19 %	7.89
	LOO_NEV	476202.42	534421.37	50.10 %	7.60
	LOO_Event	487659.19	544940.34	51.48 %	7.78
XGB	C0	495429.52	552003.11	52.37 %	7.91
	Call [†]	493905.28	545876.55	52.63 %	7.88
	LOO_Wind	492880.57	548242.76	52.25 %	7.86
	LOO_Magnet	506006.00	559534.75	53.87 %	8.07
	LOO_Production	488375.91	541417.72	51.88 %	7.79
	LOO_NEV	478495.57	532167.14	50.77 %	7.64
	LOO_Event	506098.58	559356.88	53.90 %	8.08
LGBM	C0	461763.01	520354.65	48.44 %	7.37
	Call	472473.68	529427.44	49.80 %	7.54
	LOO_Wind	472473.68	529427.44	49.80 %	7.54
	LOO_Magnet	472473.68	529427.44	49.80 %	7.54
	LOO_Production	472473.68	529427.44	49.80 %	7.54
	LOO_NEV	472473.68	529427.44	49.80 %	7.54
	LOO_Event	472473.68	529427.44	49.80 %	7.54
CAT	C0	486878.41	545232.15	51.27 %	7.77
	Call	486409.19	543343.49	51.37 %	7.76
	LOO_Wind	482290.72	539440.89	50.88 %	7.70
	LOO_Magnet	486299.23	542563.38	51.42 %	7.76
	LOO_Production	485227.40	541378.85	51.30 %	7.74
	LOO_NEV	478977.37	536880.30	50.45 %	7.64
	LOO_Event	482582.81	539305.80	50.95 %	7.70

Notes. Bold numbers indicate column-wise minima within the scenario. The [†] symbol (†) marks the author-selected primary specification based on trajectory fidelity and robustness; it need not coincide with the numerical minima. Boldface applied to names in the *Spec* column (if present) is an author highlight and does not encode statistical significance.

Table 10:

Full results for R2 (COVID→Recovery | train=*covid_impact*,
test=*recovery*).

Algorithm	Spec	MAE	RMSE	MAPE	MASE
RF	C0	167868.70	178869.40	32.67 %	0.43
	Call	350705.00	376186.61	68.12 %	0.90
	LOO_Wind	376353.70	405018.18	73.12 %	0.96
	LOO_Magnet	382412.22	413870.56	74.50 %	0.98
	LOO_Production	340077.96	365553.17	66.11 %	0.87
	LOO_NEV	347160.74	374307.75	67.57 %	0.89
	LOO_Event	150485.37	160105.70	29.09 %	0.38
XGB	C0	387189.49	466790.41	75.44 %	0.99
	Call	331560.80	379934.69	64.75 %	0.85
	LOO_Wind	405048.75	457296.62	78.92 %	1.03
	LOO_Magnet	534561.20	601359.33	104.09 %	1.36
	LOO_Production	332387.13	382263.75	64.95 %	0.85
	LOO_NEV	327170.84	378487.43	64.05 %	0.84
	LOO_Event †	78405.68	89116.18	15.14 %	0.20
LGBM	C0	396875.56	401075.42	75.56 %	1.01
	Call	396875.56	401075.42	75.56 %	1.01
	LOO_Wind	396875.56	401075.42	75.56 %	1.01
	LOO_Magnet	396875.56	401075.42	75.56 %	1.01
	LOO_Production	396875.56	401075.42	75.56 %	1.01
	LOO_NEV	396875.56	401075.42	75.56 %	1.01
	LOO_Event	396875.56	401075.42	75.56 %	1.01
CAT	C0	262754.38	277254.26	49.89 %	0.67
	Call	338478.66	350316.01	65.21 %	0.86
	LOO_Wind	360683.99	373517.56	69.49 %	0.92
	LOO_Magnet	359545.03	374487.50	69.41 %	0.92
	LOO_Production	327340.50	341289.57	63.16 %	0.84
	LOO_NEV	337989.12	352890.11	65.30 %	0.86
	LOO_Event	266239.85	273035.14	51.04 %	0.68

Notes. Bold numbers indicate column-wise minima within the scenario. The † symbol (†) marks the author-selected primary specification based on trajectory fidelity and robustness; it need not coincide with the numerical minima. Boldface applied to names in the *Spec* column (if present) is an author highlight and does not encode statistical significance.

Table 11:

Full results for R3 (Pre+COVID→Recovery | train= $pre_covid \cup covid_impact$, test= $recovery$).

Algorithm	Spec	MAE	RMSE	MAPE	MASE
RF	C0	231072.22	257700.81	45.29 %	1.31
	Call	287587.41	299315.85	55.19 %	1.64
	LOO_Wind	371147.41	392953.59	71.84 %	2.11
	LOO_Magnet	262871.48	274038.92	50.51 %	1.49
	LOO_Production	273370.19	285117.39	52.41 %	1.55
	LOO_NEV	238832.78	251273.09	46.07 %	1.36
	LOO_Event	139561.85	152390.97	26.46 %	0.79
XGB	C0	115045.42	125584.00	20.60 %	0.65
	Call	49615.73	60499.61	9.24 %	0.28
	LOO_Wind	274673.62	320707.75	53.45 %	1.56
	LOO_Magnet	74281.50	108511.76	13.78 %	0.42
	LOO_Production [†]	49912.42	59974.98	9.39 %	0.28
	LOO_NEV	72380.07	100456.03	13.43 %	0.41
	LOO_Event	58285.39	67920.13	11.46 %	0.33
LGBM	C0	444109.10	458861.53	84.96 %	2.53
	Call	183161.59	196235.22	34.99 %	1.04
	LOO_Wind	296457.26	308273.67	56.84 %	1.69
	LOO_Magnet	246489.23	260863.39	47.27 %	1.40
	LOO_Production	163713.25	181604.14	31.35 %	0.93
	LOO_NEV	159929.88	176483.86	30.68 %	0.91
	LOO_Event	196219.38	206127.01	37.15 %	1.12
CAT	C0	373319.76	422866.95	73.01 %	2.12
	Call	155832.11	165685.51	30.24 %	0.89
	LOO_Wind	209150.93	223081.04	40.67 %	1.19
	LOO_Magnet	180459.30	191483.20	34.97 %	1.03
	LOO_Production	158164.49	167586.24	30.48 %	0.90
	LOO_NEV	161218.21	170982.70	31.22 %	0.92
	LOO_Event	134560.07	142027.18	26.00 %	0.77

Notes. Bold numbers indicate column-wise minima within the scenario. The [†] symbol (†) marks the author-selected primary specification based on trajectory fidelity and robustness; it need not coincide with the numerical minima. Boldface applied to names in the *Spec* column (if present) is an author highlight and does not encode statistical significance.

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